Data Exploration

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IDS 570: Text as Data

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Preprocessing Choices

I preprocessed the texts when calculating TF-IDF and Pearson correlation values. The following are my normalization and cleaning choices, along with their justifications:

- *ſ to the modern s*: the normalization of the long S to the modern S was required for this project.
- *Collapse white space*: historical texts can have irregular spaces due to archaic typography. The irregular spacing is irrelevant to semantic, lexical, or syntactic meaning.
- *Lowercase words*: I want to analyze the frequency and context in which different tokens occur in the documents, regardless of capitalization.
- *Remove stopwords*: I removed the stopwords that are in tidytext's built-in stopword list, along with a collection of custom stopwords that are common in 17th century texts. Stopwords will add noise.
- *Remove punctuation*: I am not analyzing the style of these documents, so punctuation will likely not add value to this analysis.
- *Remove numbers*: Pagination and rounded or estimated numbers may be distracting.
- *Remove symbols*: Likely, the only meaningful symbols are currency markers. These markers will be minimally useful because all numbers have been removed.

TF-IDF: Lexical Distinctiveness

For each document, I extracted the top 15 terms with the highest TF-IDF values. The top terms are visualized in Appendix A. Then, from each document's list of top 15 TF-IDF terms, I selected the unique document terms (i.e. terms that only occurred in one document and nowhere else in the corpus). Appendix B contains a table for each set of distinctive terms per document.

Are distinctive terms topical, rhetorical, or technical?

Distinctive terms seem to be topical and technical, though I am unfamiliar with several terms that are not a part of modern vocabulary. Distinctive topical terms include: "ship," "crowns," "destruction," "commerce," and "exportations." Distinctive technical terms include: "london-bridge," "husball," and "surplusage."

Are there documents whose "distinctiveness" seems driven by noise or formatting rather than content?

No, documents' "distinctiveness" does not seem to be driven by noise or formatting, likely because of the aforementioned normalization choices.

Pearson Correlation: Similarity and Distance Between Texts

Before computing correlations, I trimmed rare words (words that occurred less than five times in a text) from the document feature matrix. The results are visualized in the similarity heatmap shown in Figure 1.1.

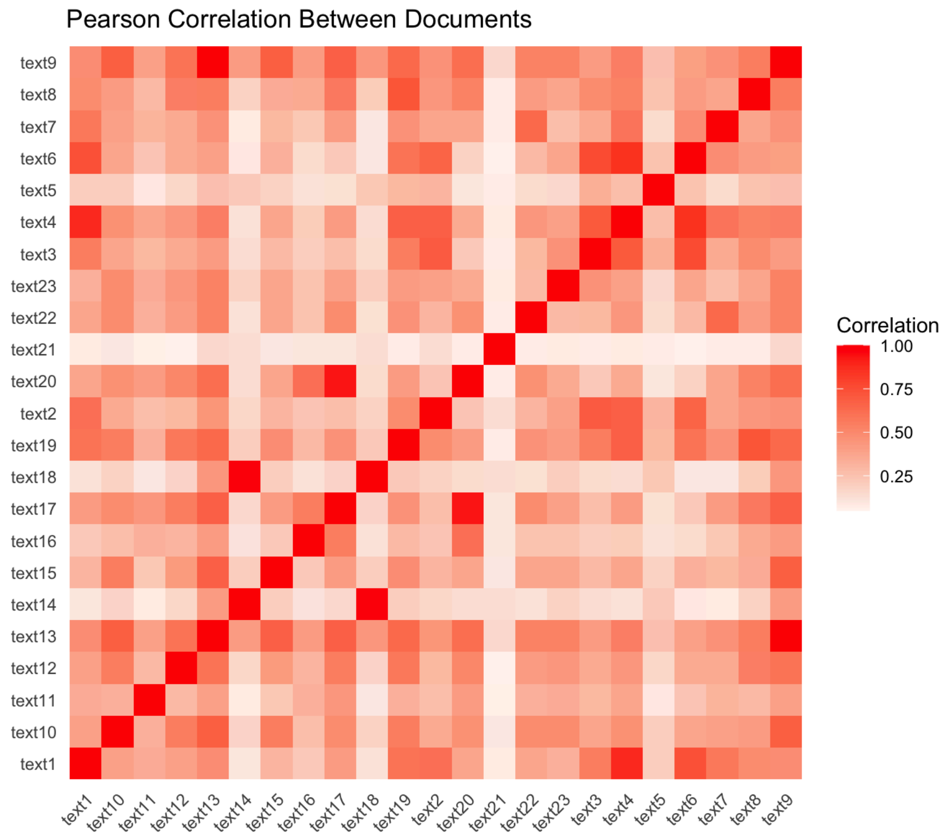


Figure 1.1: Pearson Correlation Between Documents

Identification of Most and Least Similar Document Pairs

The most similar document pair was Text 13 and Text 9 with $r = 1$. These documents are, in fact, identical. The least similar document pair was Text 12 and Text 21 with $r = 0.049$.

Questions and Analysis of Patterns

I predict that the following documents share the same author, considering the moderate to high correlation between them: Text 3, Text 4, Text 6, and Text 14. I predict that the author(s) of Text 14, Text 15, Text 18, and Text 21 have opinions or claims that are contradictory to the rest of the corpus. Based on the Pearson correlations of this corpus, I would ask the following questions:

Which words appear together?

Text 9/Text 13 seems to have the highest Pearson correlation values with other documents in the corpus. I am interested in the contextual proximity of top IF-IDF terms for this document because it would indicate which words are strongly associated with the main topical words.

Why is Text 21 weakly correlated with the other documents in the corpus?
In text analysis, outliers usually represent meaningful distinctions. As an outlier, I predict that Text 21 is from a different genre or is about a different topic than the other documents in this corpus.

Syntactic Complexity Profile

I evaluated the syntactic complexity of the two documents with the lowest Pearson correlation coefficient: text 12 (“A32830.txt”) and text 21 (“A93819”). I chose these texts because I want to compare how these texts organize meaning; the results may help to explain their low Pearson correlation coefficient. The syntactic complexity measurements are shown in Table 2.1 and Figure 2.1 below. The sentence lengths in each document are comparable, but the other dimensions differ.

Document	MLS	C/S	DC/C	DC/S	Coord/C	Coord/S	CN/C	CN/S
text 12	41.58	3.68	0.89	3.27	1.00	3.68	1.44	5.31
text 21	43.62	2.67	0.84	2.25	2.32	6.18	2.83	7.55

Table 2.1: Syntactic Complexity Profiles of Text 12 and Text 21

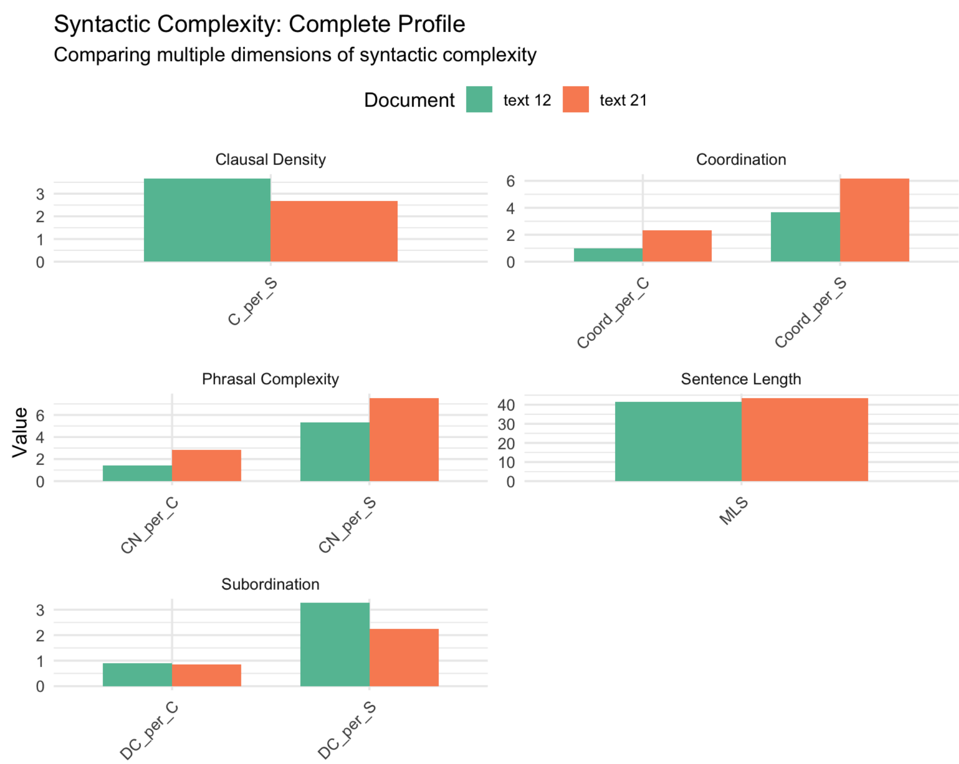


Figure 2.1: Visualization of the Syntactic Complexity Profiles of Text 12 and Text 21

How do the two texts differ in syntactic complexity?
The sentence length between the two texts is comparable; the mean length of sentence differs by 2.04 words. Additionally, the average amount of subordination per clause is similar between documents. Overall, Text 21 is more syntactically complex than Text 12. Text 21 has higher coordination and more complex nominals.

Coordination
For this measurement, I counted *any* coordination marker. In other words, I am counting not only the number of coordinated clauses, but rather the frequency of parallel syntactic structures.

I highlighted the coordination markers in the first sentence of Text 12:

> IT cannot but seem something strange that it should be thought necessary in this Age of ours, that pretends to see so much further than their Ancestors: for a Man to give himself any trouble to Oppose an Assertion, which in the mildest Construction, cannot pass for less than a Paradox.

Similarly, I highlighted the coordination markers in the first sentence of Text 21:

> CSTHe said Persons in the Name of the City of London, (but for their own private advantage only) have Indicted the Defendant for stopping up the Stairs at Lion-Key aforesaid, and a passage formerly leading down to them; and for planking over eight foot into the River of Thames, and for erecting a Crane upon the said Key, which they say are public Nuisances.

These example sentences support the measurements above. Indeed, there were three coordination markers in the first sentence of Text 12, and Text 12 on average has 3.68 coordination markers per sentence. There were eight coordination markers in the first sentence of Text 21, and Text 21 on average has 6.18 coordination markers per sentence.

Do these differences align with or complicate the lexical findings?
These differences align with the lexical findings. The low Pearson correlation score suggests that the documents are structured differently, and indeed, they are syntactically different. I predict that the syntactic complexity of other documents in this corpus are more similar to the measurements of Text 12 than Text 21.

What kinds of rhetorical or stylistic practices might these syntactic patterns reflect?
These syntactic patterns reflect an academic writing style. Likely, only wealthy, literate white males were invited to study and comment on political economies. The syntactic complexity makes the texts difficult to read, so a formal education was likely a prerequisite to reading and understanding these texts. I predict that Text 21 was written for an even more elite audience than the other texts, based on its higher syntactic complexity.

Synthesis

I have not read the texts about which this report is written, but I know that they are somehow related by the theme "political economies." Upon completing this exploratory text analysis, I formulated this research question: *What commodities were commonly exchanged during this time period, and which were of greatest value?*

Some of the tokens with high TF-IDF scores can be categorized as commodities, measurements, or currencies. TF-IDF accentuates terms that are distinctly frequent in a document, but not frequent in the corpus, overall. TF is greater when the word is frequent in the document, and IDF is greater when the term is rare in the corpus (and may be unique to a document). Based on documents' top TF-IDF terms, I infer that the documents discuss trading. I assume that commodities had varying values, and a more comprehensive analysis could reveal which commodities were most commonly traded, and which were most highly valued.

Commodities	**Measurements**	**Currencies**
commodities cloth grains indigo raw-silkes stivers velvet wool	carats ounces pieces weight	cent crowns doller ducats gold moneys silver sterling

Table 3.1: Tokens from Documents' Top TF-IDF Terms (see Appendix A), Organized into Three Categories: "Commodities," "Measurements," and "Currencies"

I filtered the top 15 TF-IDF terms across all documents (345 terms in total) to those that occur in at least five of the 23 documents. Only three tokens —"realm," "moneys," and "india" — were present in five or more documents. Interestingly, six of the seven texts that contained "realm" also contained "moneys" but not "india." Similarly, six of the seven texts that contained "moneys" also contained "realm" but not "india." This suggests that specific currencies (rather than the general term "moneys") were used to discuss trade in India. Also, the words "moneys" and "realm" may share semantic meaning.

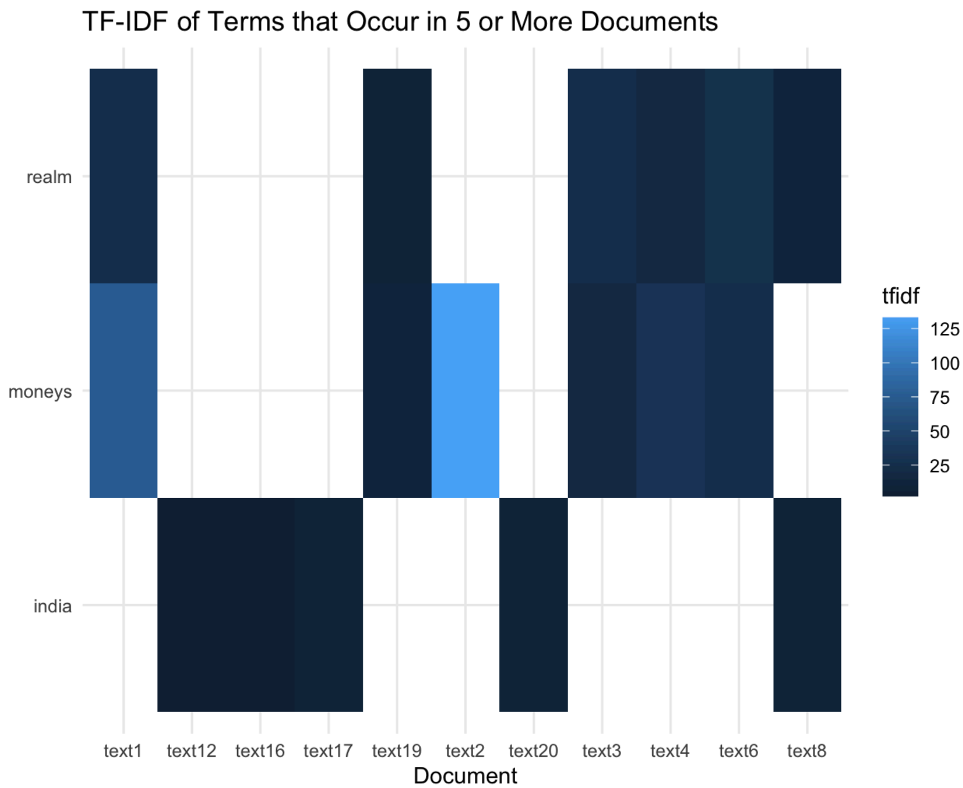


Figure 3.1: TF-IDF of Tokens that Occur in 5 or more Documents

Based on the heatmap of Pearson correlation between documents(see Figure 1.1), Text 3, Text 4, Text 6, and Text 14 are moderately to highly correlated. I also took a closer look at the data and identified the top ten document pairs with the highest Pearson correlation coefficients (see Table 3.2). Text 14 and Text 18 have an *r* value of 0.979 and the same top three TF-IDF terms, “poor,” “parish,” and “fathers.” Text 4 and Text 1 have an *r* value of 0.877 and share two of the three top TF-IDF terms. Although Pearson correlation does not assume that shared vocabulary implies similarity on its own, this data suggests that the same vocabulary is of relatively equal distinctiveness across pairwise documents. Text 6 and Text 4 have an *r* value of 0.858, and the top TF-IDF terms are commodities, measurements, or currencies. Thus, I predict that reading these texts will reveal answers to my research question.

	doc_i	text9	text18	text20	text4	text6	text8	text3
1	text13	1	NA	NA	NA	NA	NA	NA
2	text14	NA	0.979	NA	NA	NA	NA	NA
3	text17	NA	NA	0.931	NA	NA	NA	NA
4	text1	NA	NA	NA	0.877	0.746	NA	NA
5	text4	NA	NA	NA	NA	0.858	NA	NA
6	text3	NA	NA	NA	0.703	0.770	NA	NA
7	text19	NA	NA	NA	NA	NA	0.728	NA
8	text2	NA	NA	NA	NA	NA	NA	0.692

Table 3.2: Top 10 Document Pairs with the Highest Pearson Correlation Coefficients

Finally, I calculated syntactic complexity values for all documents in the corpus (see Table 3.3). The document pairs with high Pearson correlation coefficients generally have similar syntactic complexities. Syntactic complexity could indicate authors' higher levels of education. Perhaps, documents with high syntactic complexity contain more nuanced data and analyses about traded commodities and their values.

Syntactic Complexity of All Documents in the Corpus

	document	MLS	C_per_S	DC_per_C	DC_per_S	Coord_per_C	Coord_per_S	CN_per_C	CN_per_S
1	text 1	31.136	2.143	0.951	2.038	1.612	3.454	2.233	4.786
12	text 2	35.334	2.265	1.012	2.293	2.150	4.869	2.247	5.089
17	text 3	32.773	2.189	1.002	2.192	1.817	3.978	2.147	4.700
18	text 4	27.089	1.688	1.047	1.769	1.732	2.924	2.499	4.220
19	text 5	44.665	2.961	1.237	3.661	1.857	5.498	1.972	5.840
20	text 6	43.009	3.021	0.963	2.910	1.720	5.195	1.922	5.807
21	text 7	21.234	1.603	0.695	1.115	1.577	2.527	2.258	3.618
22	text 8	27.998	1.903	0.897	1.708	1.660	3.159	2.333	4.441
23	text 9	41.680	3.381	0.862	2.913	1.490	5.038	1.702	5.754
2	text 10	44.767	3.186	0.964	3.070	1.978	6.302	1.915	6.101
3	text 11	35.074	2.680	0.884	2.369	1.685	4.516	1.789	4.795
4	text 12	41.577	3.676	0.889	3.268	1.000	3.676	1.444	5.310
5	text 13	40.963	3.323	0.860	2.858	1.494	4.965	1.715	5.697
6	text 14	41.474	3.362	0.936	3.147	1.464	4.922	1.636	5.500
7	text 15	26.449	2.390	0.784	1.873	1.092	2.610	1.305	3.119
8	text 16	35.466	2.205	0.881	1.943	1.964	4.330	2.619	5.773
9	text 17	27.848	2.276	0.730	1.661	1.500	3.415	1.751	3.986
10	text 18	42.080	3.272	0.944	3.088	1.589	5.200	1.773	5.800
11	text 19	47.638	3.791	0.952	3.610	1.523	5.776	1.578	5.983
13	text 20	22.855	1.721	0.678	1.168	1.532	2.637	2.204	3.794
14	text 21	43.617	2.667	0.844	2.250	2.319	6.183	2.831	7.550
15	text 22	23.575	1.561	0.761	1.189	1.770	2.764	2.692	4.203
16	text 23	32.068	2.326	0.822	1.912	1.263	2.937	2.240	5.207

Table 3.3: Syntactic Complexity of All Documents in the Corpus

Overall, three quantitative approaches —TF-IDF, Pearson correlation, and syntactic complexity measures— helped me identify topics in the corpus and similarities between documents. However, semantic analysis is necessary to understand the content of the text and answer the research question.

Appendix A
Top 15 TF-IDF Terms by Document

Figure A1

Top 15 TF-IDF Terms in Text 1

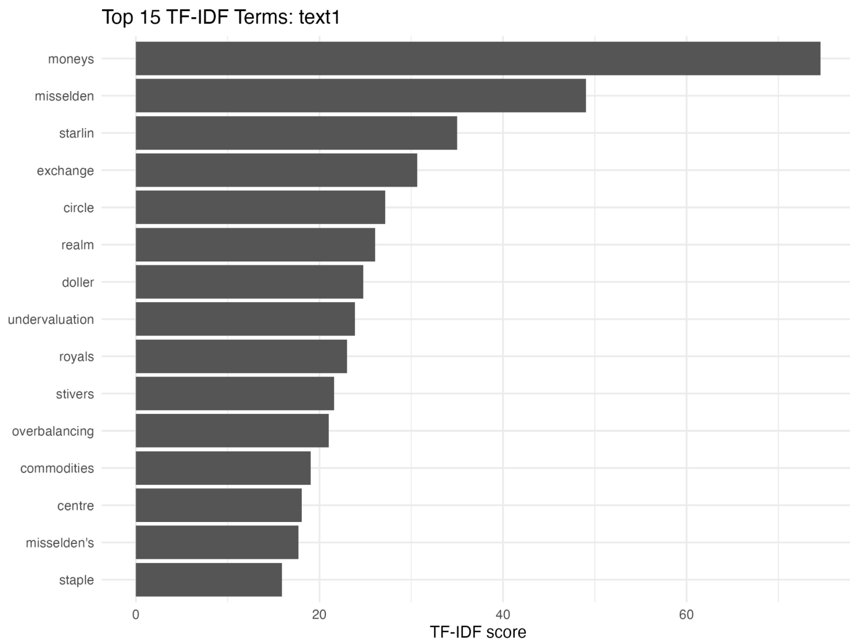


Figure A2

Top 15 TF-IDF Terms in Text 2

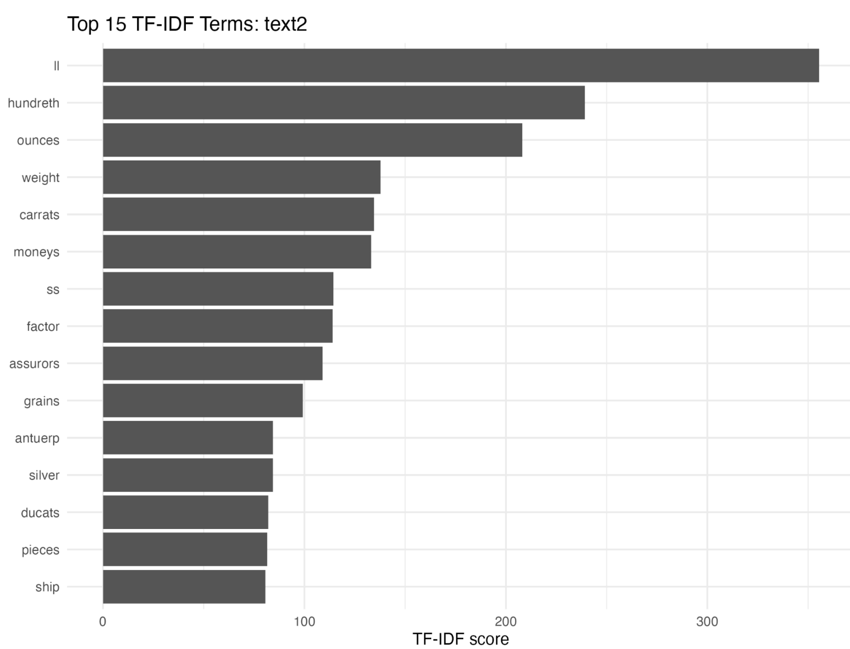


Figure A3

Top 15 TF-IDF Terms in Text 3

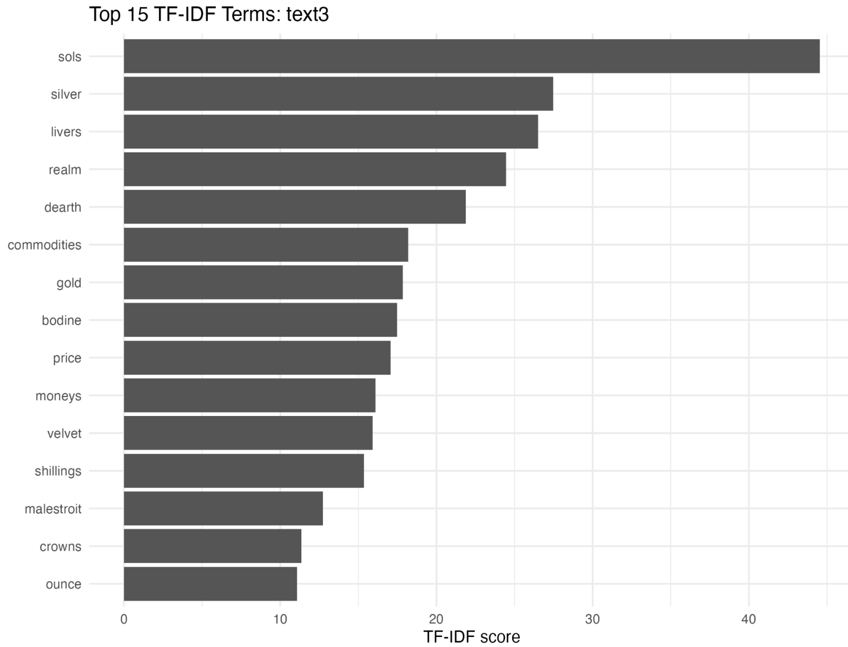


Figure A4

Top 15 TF-IDF Terms in Text 4

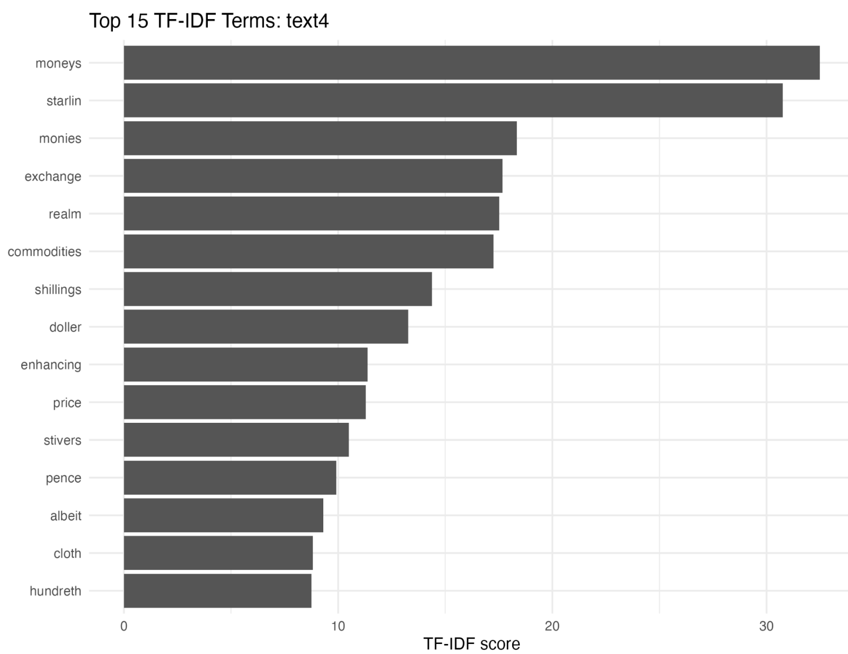


Figure A5

Top 15 TF-IDF Terms in Text 5

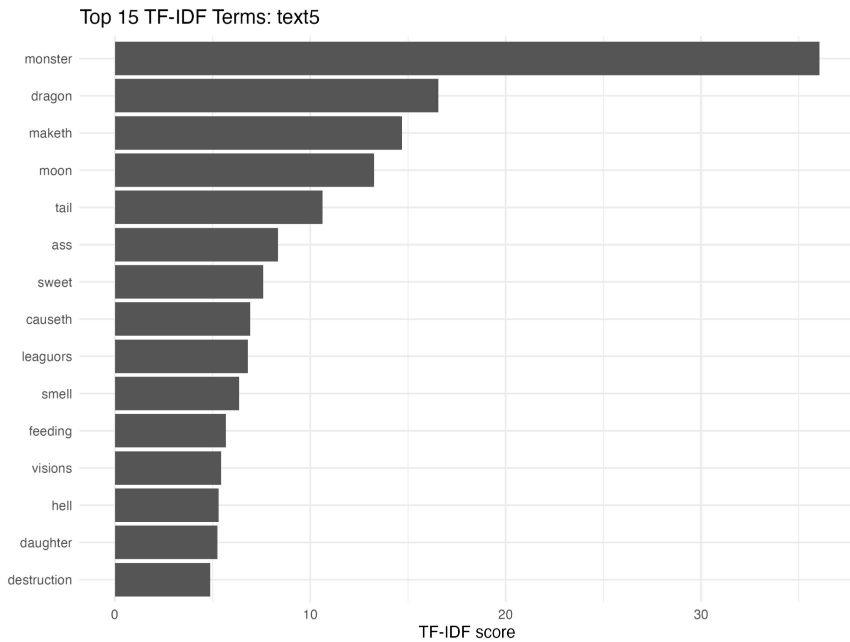


Figure A6

Top 15 TF-IDF Terms in Text 6

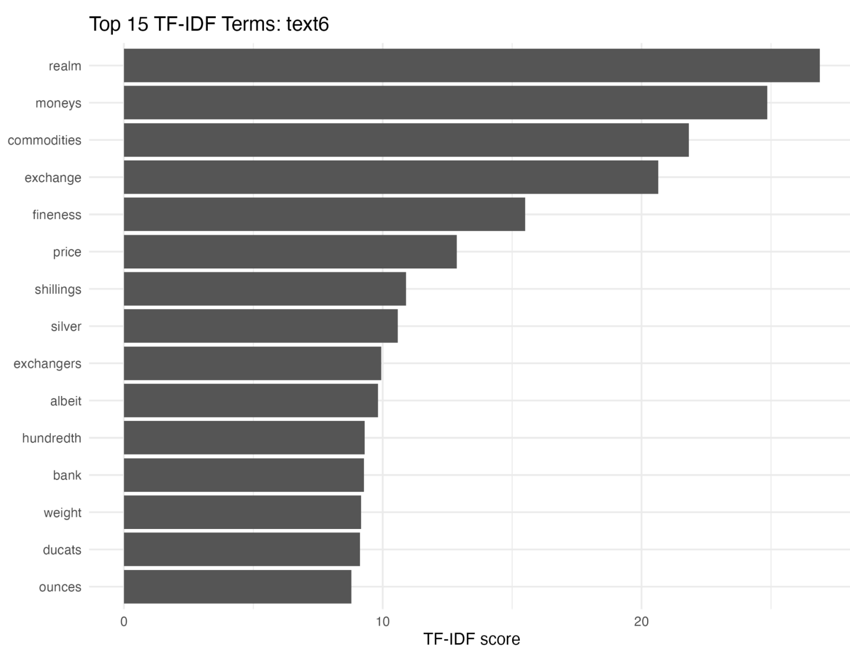


Figure A7

Top 15 TF-IDF Terms in Text 7

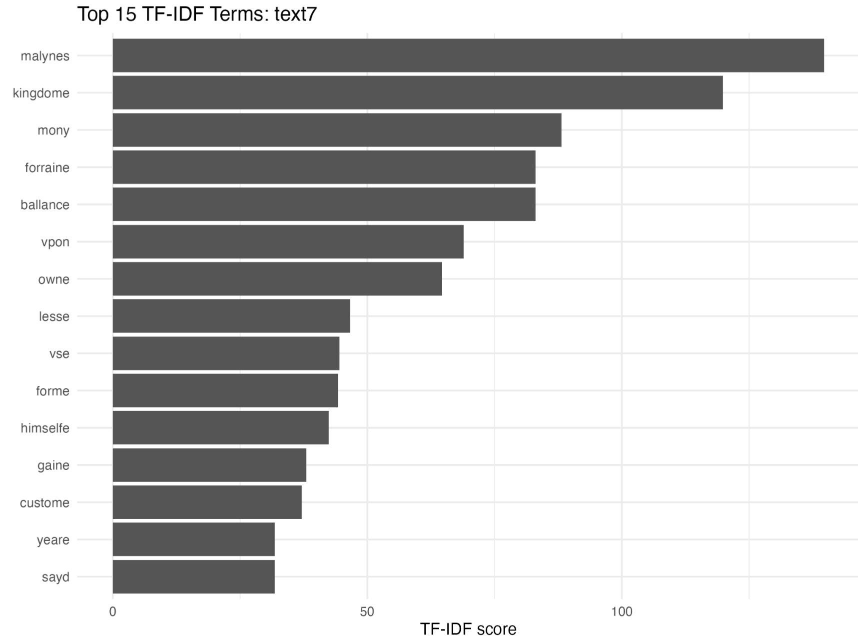


Figure A8

Top 15 TF-IDF Terms in Text 8

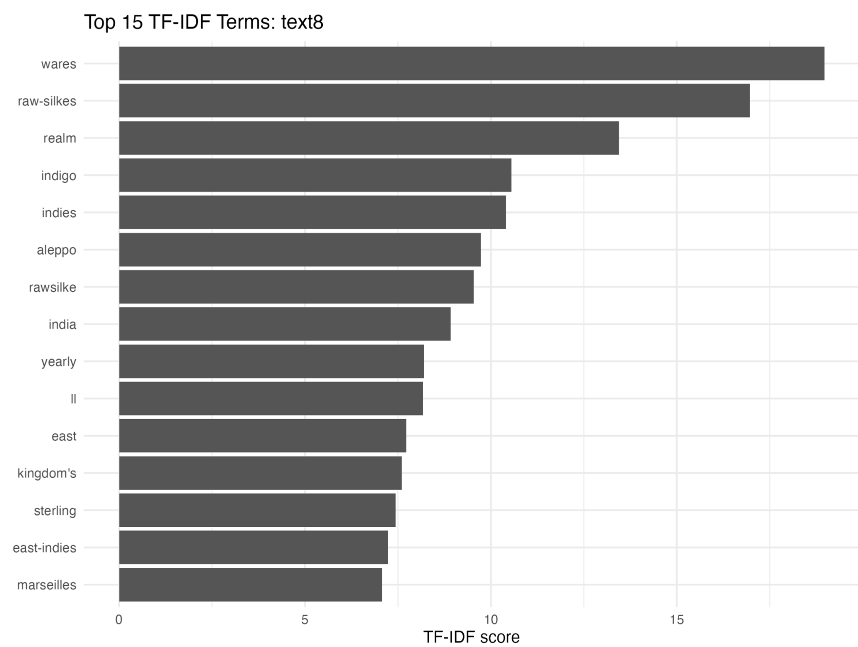


Figure A9

Top 15 TF-IDF Terms in Text 9

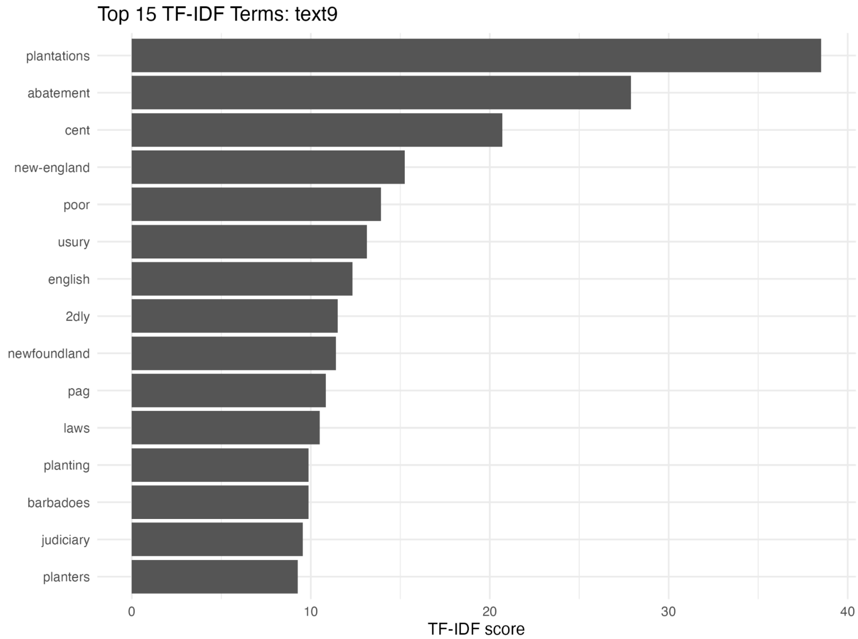


Figure A10

Top 15 TF-IDF Terms in Text 10

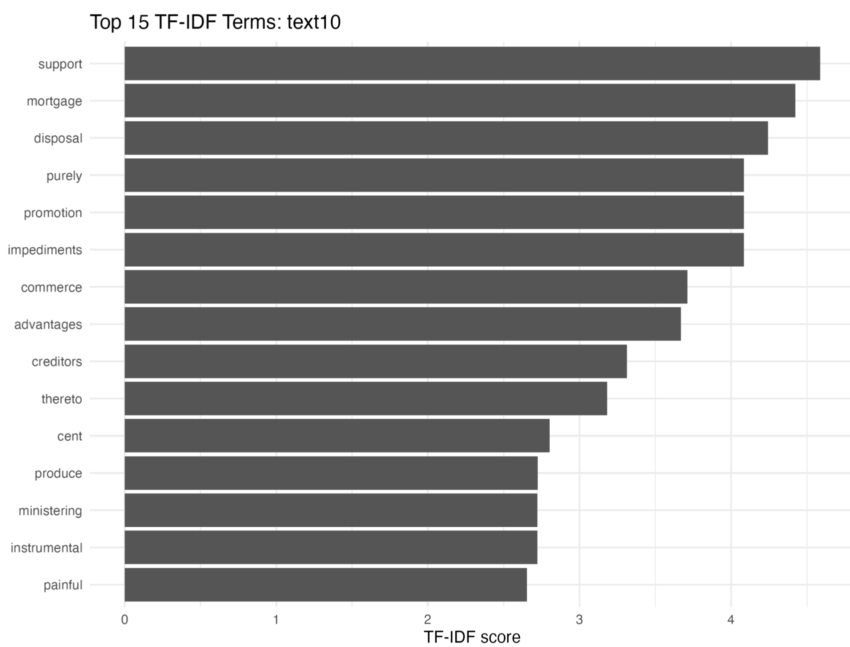


Figure A11

Top 15 TF-IDF Terms in Text 11

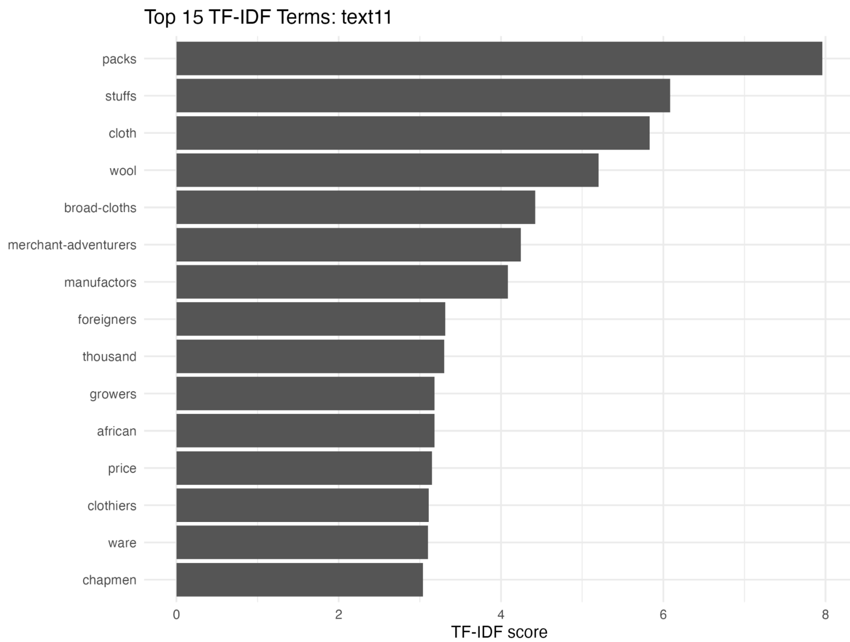


Figure A12

Top 15 TF-IDF Terms in Text 12

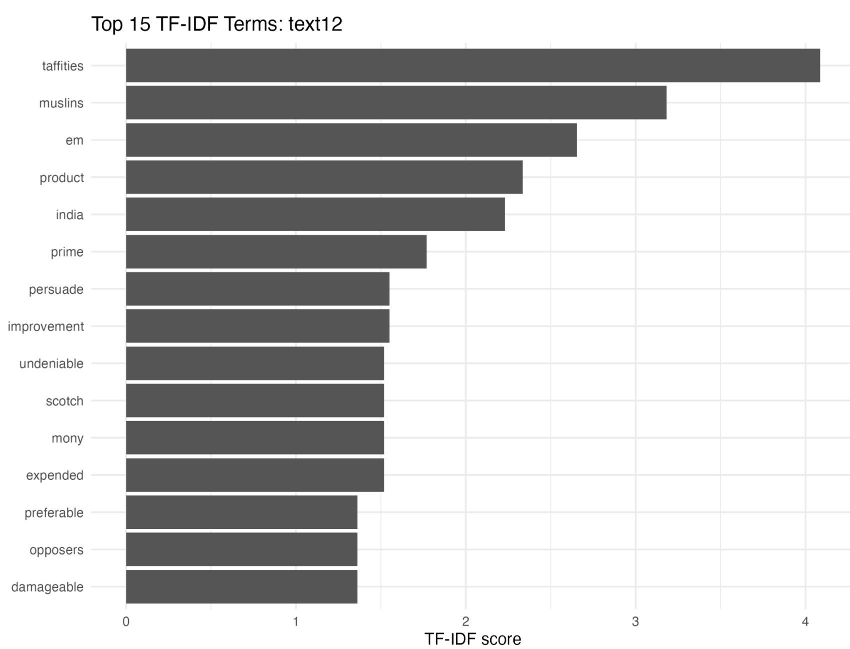


Figure A13

Top 15 TF-IDF Terms in Text 13

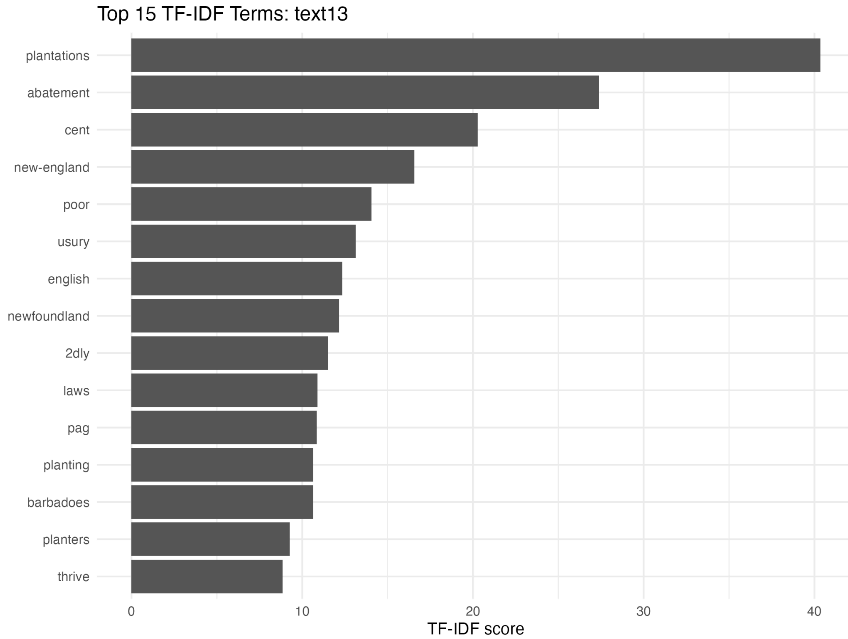


Figure A14

Top 15 TF-IDF Terms in Text 14

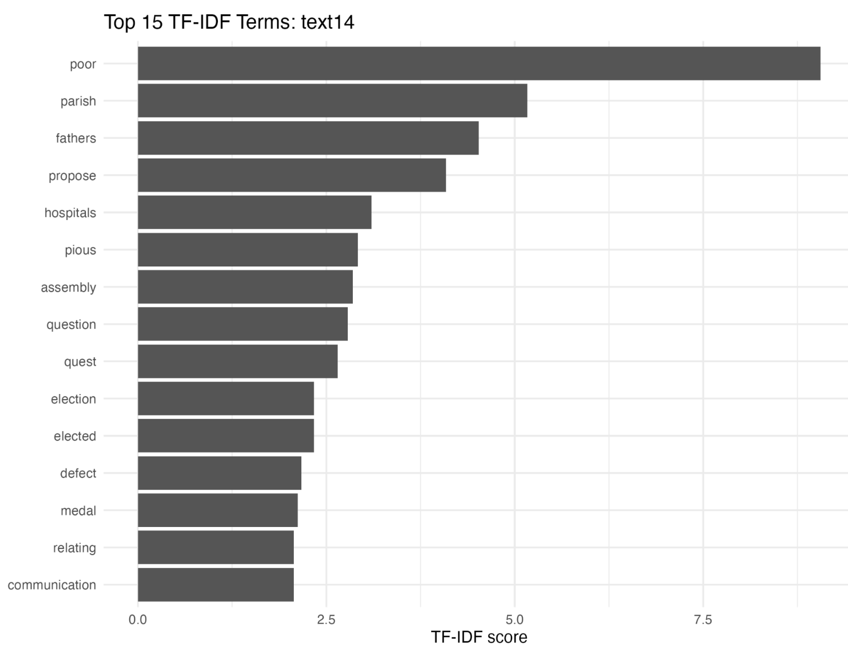


Figure A15

Top 15 TF-IDF Terms in Text 15

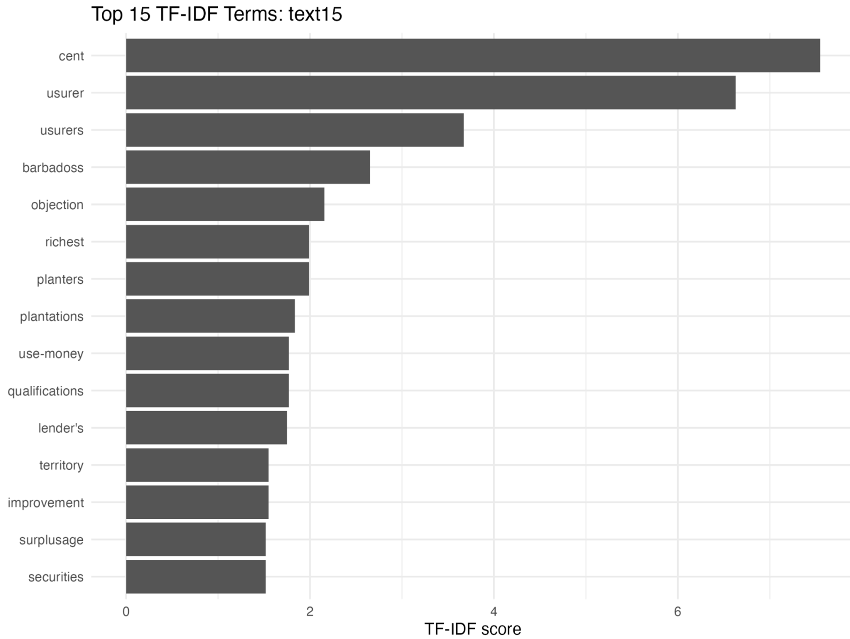


Figure A16

Top 15 TF-IDF Terms in Text 16

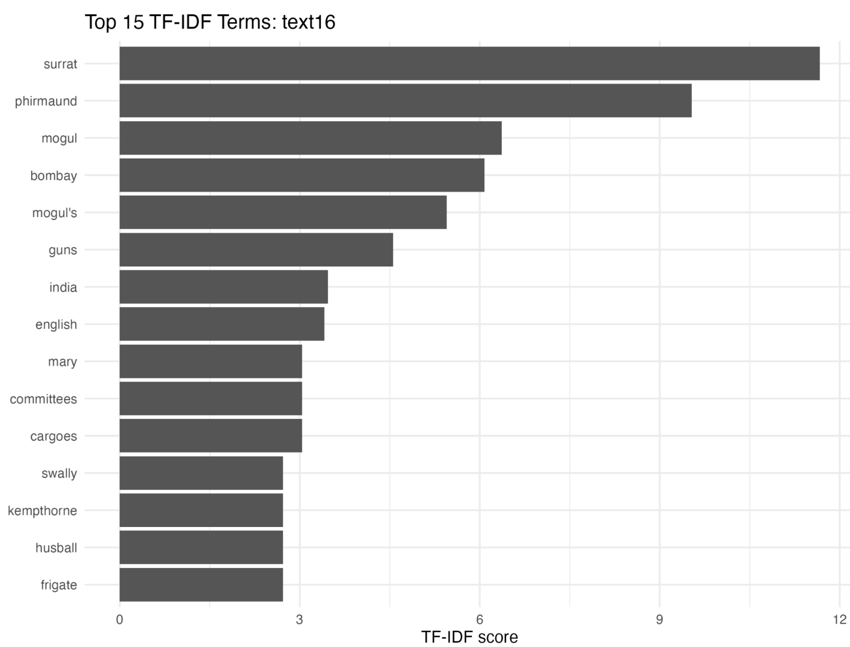


Figure A17

Top 15 TF-IDF Terms in Text 17

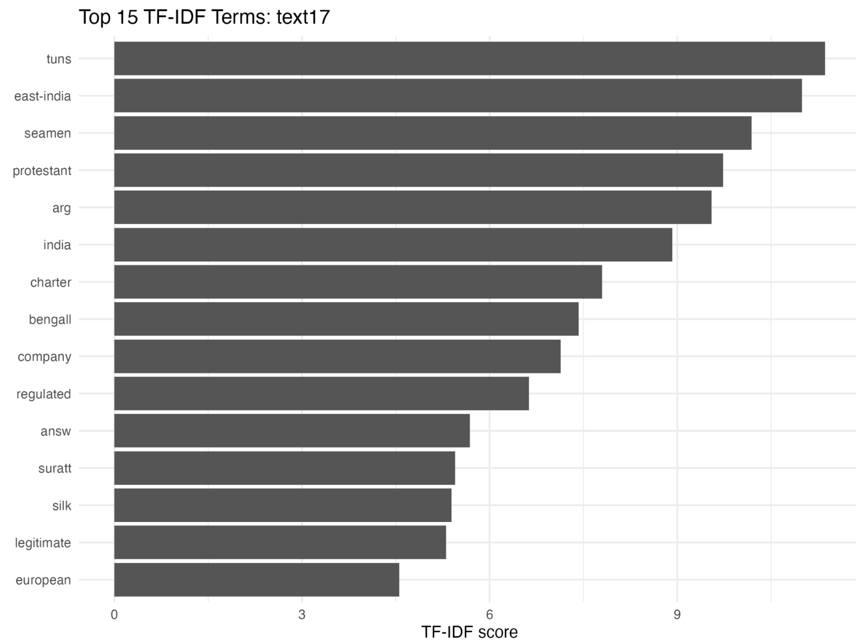


Figure A18

Top 15 TF-IDF Terms in Text 18

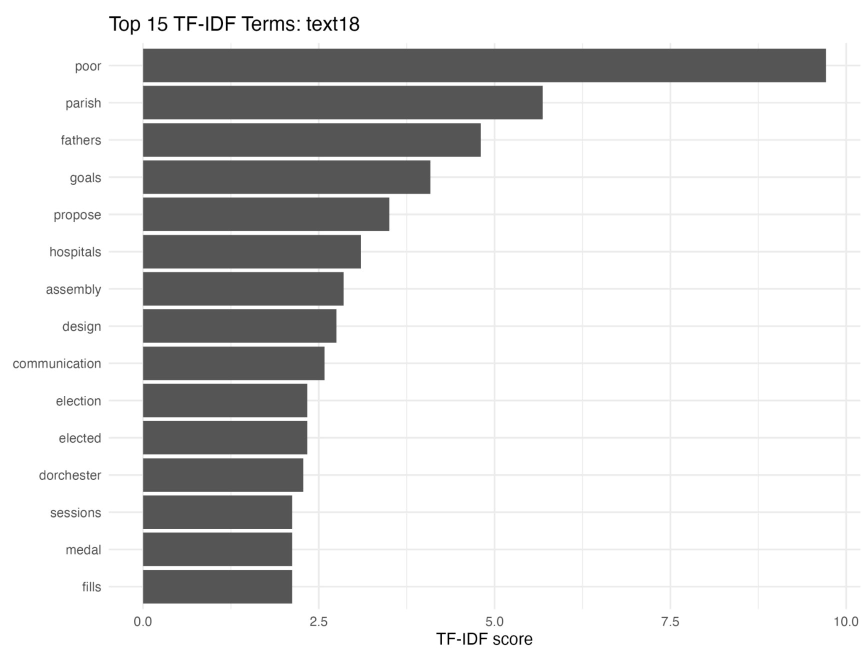


Figure A19

Top 15 TF-IDF Terms in Text 19

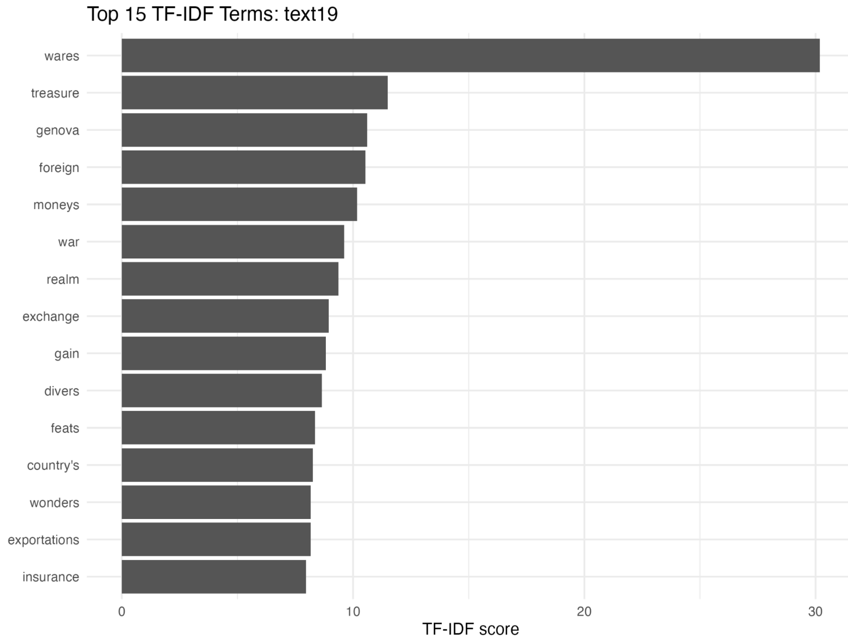


Figure A20

Top 15 TF-IDF Terms in Text 20

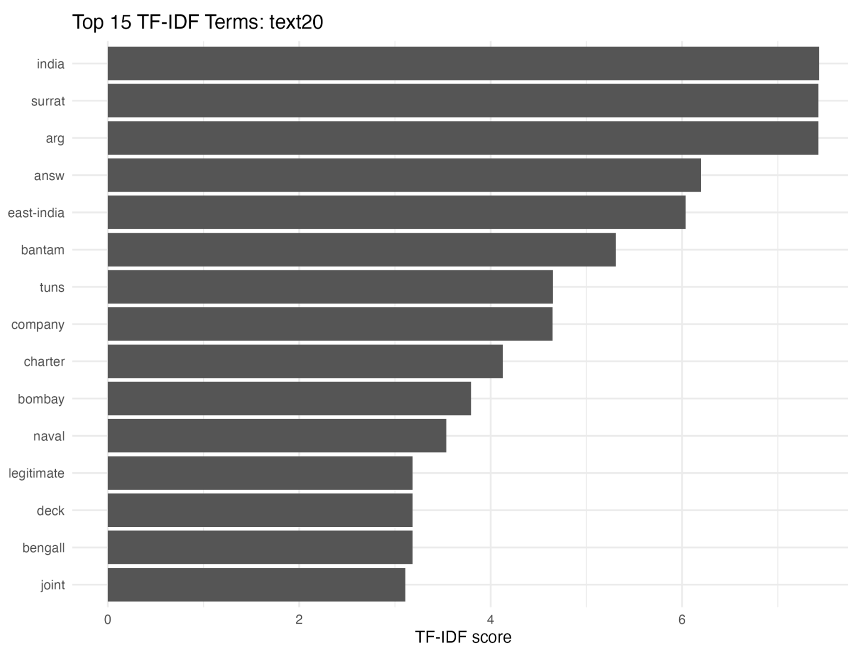


Figure A21

Top 15 TF-IDF Terms in Text 21

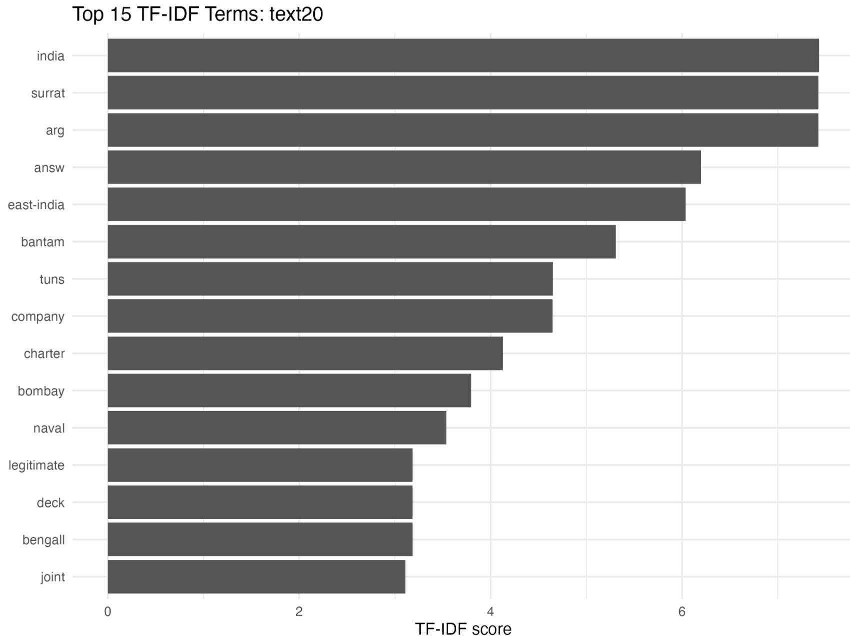


Figure A22

Top 15 TF-IDF Terms in Text 22

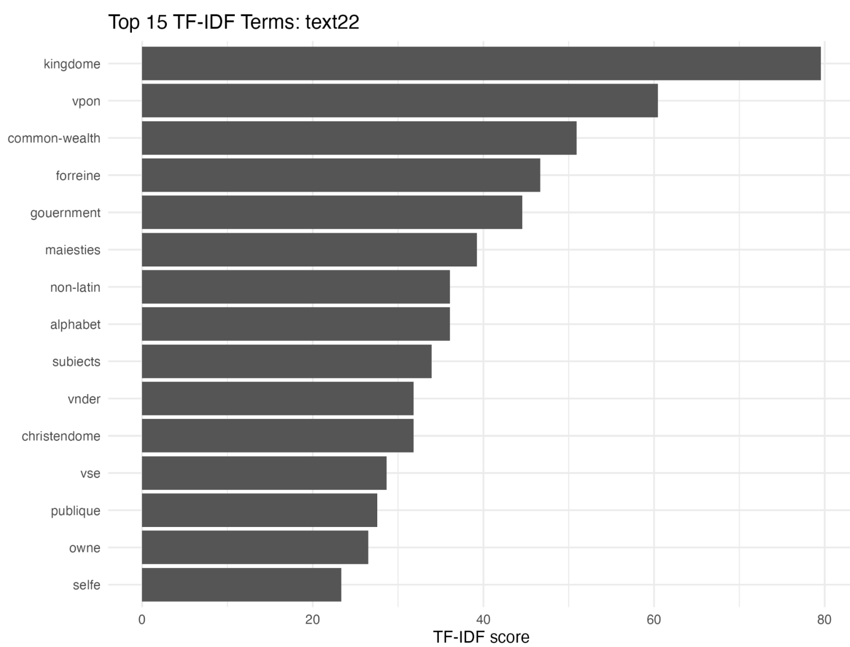
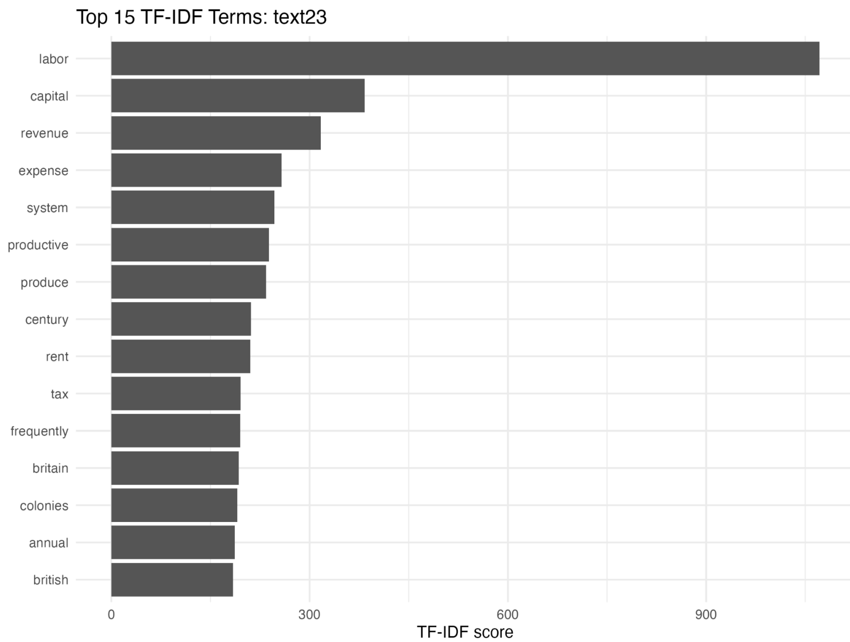


Figure A23

Top 15 TF-IDF Terms in Text 23



Appendix B
Distinctive Terms by Document

Table A1

Distinct Terms in Text 1

doc_title	word	tfidf
text1	misselden	49.0222020966333
text1	circle	27.1730710989445
text1	undervaluation	23.8592819405367
text1	royals	22.9997711137462
text1	overbalancing	21.0087570828222
text1	centre	18.082981758078
text1	misselden's	17.7024618682287
text1	staple	15.9160247847741

Table A2

Distinct Terms in Text 2

doc_title	word	tfidf
text2	carrats	134.460200357285
text2	ss	114.385138225478
text2	factor	113.994347049231
text2	assurors	108.938226881407
text2	grains	99.2080195577714
text2	antuerp	84.4271258330908
text2	pieces	81.4967245282761
text2	ship	80.6343253269187

Table A3
Distinct Terms in Text 3

doc_title	word	tfidf
text3	sols	44.5493092948517
text3	livers	26.5174460088403
text3	dearth	21.8710084454919
text3	gold	17.8539323909474
text3	bodine	17.4723604278615
text3	velvet	15.9104676053042
text3	malestroit	12.7283740842433
text3	crowns	11.3658555120734
text3	ounce	11.087955127045

Table A4
Distinct Terms in Text 4

doc_title	word	tfidf
text4	monies	18.345513961026
text4	enhancing	11.3658555120734
text4	pence	9.91137934843024

Table A5
Distinct Terms in Text 5

doc_title	word	tfidf
text5	monster	36.0637265720228
text5	dragon	16.5689457920394
text5	maketh	14.6924226784383
text5	moon	13.269098719469
text5	tail	10.6353498256548
text5	ass	8.35634629158593
text5	sweet	7.5966784468963
text5	causeth	6.93796554390117
text5	leaguors	6.80863918008796
text5	smell	6.36418704212167
text5	feeding	5.6829277560367
text5	visions	5.44691134407037
text5	hell	5.30348920176806
text5	daughter	5.25218927070554
text5	destruction	4.88982391893922

Table A6
Distinct Terms in Text 6

doc_title	word	tfidf
text6	fineness	15.4988938801001
text6	exchangers	9.94136747522361
text6	hundredth	9.29933632806005
text6	bank	9.27860964354204

Table A7
Distinct Terms in Text 7

doc_title	word	tfidf
text7	malynes	139.767839845073
text7	ballance	83.0653979970732
text7	forraine	83.0653979970732
text7	lesse	46.6707049755589
text7	forme	44.2303290648965
text7	himselfe	42.4279136141445
text7	gaine	38.038082995811
text7	custome	37.1244244123764
text7	sayd	31.8209352106083
text7	yeare	31.8209352106083

Table A8
Distinct Terms in Text 8

doc_title	word	tfidf
text8	raw-silkes	16.9711654456578
text8	indigo	10.5486705275899
text8	indies	10.4069483158518
text8	aleppo	9.73067239427724
text8	rawsilke	9.53209485212315
text8	yearly	8.19560837480674
text8	east	7.72278481472174
text8	kingdom's	7.5966784468963
text8	sterling	7.43353451132268
text8	east-indies	7.23455672035186
text8	marseilles	7.07685265038344

Table A9

Distinct Terms in Text 9

doc_title	word	tfidf
text9	judiciary	9.5462805631825

Table A10

Distinct Terms in Text 10

doc_title	word	tfidf
text10	support	4.58637849025649
text10	mortgage	4.42303290648965
text10	disposal	4.24279136141445
text10	purely	4.08518350805278
text10	promotion	4.08518350805278
text10	impediments	4.08518350805278
text10	commerce	3.71273153923823
text10	advantages	3.66910279220519
text10	creditors	3.31378915840787
text10	thereto	3.18209352106083
text10	ministering	2.72345567203519
text10	instrumental	2.72345567203519
text10	painful	2.65381974389379

Table A11
Distinct Terms in Text 11

doc_title	word	tfidf
text11	packs	7.96145923168137
text11	stuffs	6.08636786632799
text11	wool	5.20347415792588
text11	broad-cloths	4.42303290648965
text11	merchant-adventurers	4.24279136141445
text11	manufactors	4.08518350805278
text11	foreigners	3.31378915840787
text11	thousand	3.30011525834289
text11	growers	3.18209352106083
text11	african	3.18209352106083
text11	clothiers	3.10801248966965
text11	ware	3.09977877602002
text11	chapmen	3.03867137875852

Table A12
Distinct Terms in Text 12

doc_title	word	tfidf
text12	taffities	4.08518350805278
text12	muslins	3.18209352106083
text12	em	2.65381974389379
text12	product	2.3343063425358
text12	prime	1.76921316259586
text12	persuade	1.54988938801001
text12	expended	1.51933568937926
text12	scotch	1.51933568937926
text12	undeniable	1.51933568937926
text12	preferable	1.36172783601759
text12	opposers	1.36172783601759
text12	damageable	1.36172783601759

Table A13
Distinct Terms in Text 13

doc_title	word	tfidf
text13	thrive	8.8460658129793

Table A14
Distinct Terms in Text 14

doc_title	word	tfidf
text14	pious	2.91788292816975
text14	question	2.78454865442867
text14	quest	2.6510313267263
text14	defect	2.17036701610556
text14	relating	2.06651918401334

Table A15
Distinct Terms in Text 15

doc_title	word	tfidf
text15	usurer	6.62757831681574
text15	usurers	3.66910279220519
text15	barbadoss	2.65381974389379
text15	objection	2.15599800339355
text15	richest	1.98827349504472
text15	use-money	1.76921316259586
text15	qualifications	1.76921316259586
text15	lender's	1.75072975690185
text15	territory	1.54988938801001
text15	surplusage	1.51933568937926
text15	securities	1.51933568937926

Table A16
Distinct Terms in Text 16

doc_title	word	tfidf
text16	phirmaund	9.53209485212315
text16	mogul	6.36418704212167
text16	mogul's	5.44691134407037
text16	guns	4.55800706813778
text16	mary	3.03867137875852
text16	committees	3.03867137875852
text16	cargoes	3.03867137875852
text16	kempthorne	2.72345567203519
text16	frigate	2.72345567203519
text16	swally	2.72345567203519
text16	husball	2.72345567203519

Table A17
Distinct Terms in Text 17

doc_title	word	tfidf
text17	seamen	10.1871331644567
text17	protestant	9.73067239427724
text17	regulated	6.62757831681574
text17	suratt	5.44691134407037
text17	silk	5.38999500848387
text17	european	4.55800706813778

Table A18
Distinct Terms in Text 18

doc_title	word	tfidf
text18	goals	4.08518350805278
text18	design	2.7518270941539
text18	dorchester	2.27900353406889
text18	sessions	2.12139568070722
text18	fills	2.12139568070722

Table A19
Distinct Terms in Text 19

doc_title	word	tfidf
text19	treasure	11.497175738743
text19	genova	10.6069784035361
text19	foreign	10.5390777605144
text19	war	9.60658405897891
text19	gain	8.82603978825341
text19	divers	8.64904907320293
text19	feats	8.35634629158593
text19	country's	8.26607673605338
text19	wonders	8.17036701610556
text19	exportations	8.17007219887529
text19	insurance	7.96145923168137

Table A20
Distinct Terms in Text 20

doc_title	word	tfidf
text20	bantam	5.30763948778758
text20	naval	3.53842632519172
text20	deck	3.18209352106083
text20	joint	3.10801248966965

Table A21

Distinct Terms in Text 21

doc_title	word	tfidf
text21	lion-key	23.1493732122991
text21	stairs	15.9530247384822
text21	edmond	8.17036701610556
text21	wiseman	7.96145923168137
text21	key	6.83701060220667
text21	defendant	5.50365418830779
text21	wharfige	5.44691134407037
text21	london-bridge	5.44691134407037
text21	surveyors	5.44691134407037
text21	thames	5.31767491282741
text21	enrolled	5.30763948778758
text21	wharves	5.30763948778758
text21	wharf	5.30348920176806
text21	keys	4.55800706813778
text21	defendants	4.24279136141445

Table A22

Distinct Terms in Text 22

doc_title	word	tfidf
text22	common-wealth	50.9134963369734
text22	forreine	46.6707049755589
text22	gouernment	44.5493092948517
text22	maiesties	39.2458200930836
text22	non-latin	36.0637265720228
text22	alphabet	36.0637265720228
text22	subiects	33.9423308913156
text22	vnder	31.8209352106083
text22	christendome	31.8209352106083
text22	publique	27.5781438491939
text22	selfe	23.3353524877795

Table A23
Distinct Terms in Text 23

doc_title	word	tfidf
text23	labor	1071.30481875715
text23	capital	383.409816761505
text23	revenue	316.918753676724
text23	expense	257.295833303389
text23	system	246.472738319184
text23	productive	238.302371303079
text23	century	211.067814582727
text23	rent	210.056134853747
text23	tax	195.592956757569
text23	frequently	194.769433093258
text23	britain	192.702913909244
text23	colonies	190.829543502301
text23	annual	186.744507402864
text23	british	183.833257862375